

280 megawatts of power; which some sustainability reports claim can power more than a quarter million households! Epoch AI's chart also shows that the power requirements of leading AI supercomputers are doubling every 13 months.

Let us take a quick look at OpenAI's ChatGPT lest they feel offended. According to reports on the internet, it took approximately 1287 megawatt-hours of electricity to train GPT-3. The daily operation of ChatGPT is estimated to consume around 39.98 million kilowatt-hours. Research indicates that an average interaction with GPT-3, such as a short conversation with the user or drafting a 100-word email, may use around 500ml water to cool the data centre.

We are, arguably, heading towards an apocalypse, thanks to AI technology!

Taking the bull by its two horns

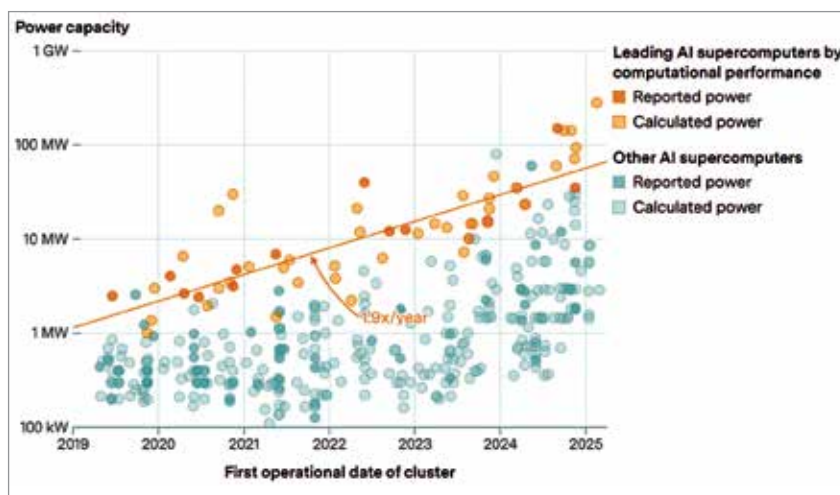
Now, there is a two-pronged problem that the AI industry worldwide is trying to solve. The first challenge is to reduce the dependency on Nvidia. Let us face the reality—many do not have access to it, some cannot afford it, and others do not want to wait in line! In response, some companies are working on developing customised GPUs for their specific needs. And some countries, including India, are encouraging the development of indigenous GPUs.

The second challenge is to find means to reduce the amount of computing power required by AI models—either work without GPUs, like Kompact AI, or reduce the number of GPUs used, like DeepSeek. This will have a positive economic and environmental impact.

Now, let us take a top-level look at a few ongoing efforts from around the world that attempt to address one or both of the above challenges.

Tech majors developing in-house AI chips

One of the root causes of the existing conundrum is the dependence of



Power consumption by leading AI supercomputers (Courtesy: Epoch AI)

almost every major AI company on one company's chips. To get out of this trap, they are investing heavily on developing their own custom AI chips, which, apart from reducing their reliance on Nvidia, will also give greater control, efficiency, and cost-effectiveness for their specific AI workloads. There is a shift towards vertical integration and application-specific acceleration, tailored for hyperscale environments and proprietary AI stacks.

Google's tensor processing units (TPUs): Google Cloud TPUs are AI accelerators optimised for training and inference of AI models. They feature special components, such as matrix multiply units (MXU) and proprietary interconnect topologies, which enable them to handle large machine learning (ML) workloads with significant cost and performance-per-watt (PPW) advantages. Hundreds or even thousands of TPUs (depending on the version) can be connected using ultra-fast, specialised networks, to make TPU pods. TPUs power Google's AI-based applications, including Gemini, Search, Photos, Maps, and more significantly, AlphaFold, an AI tool that predicts protein structures, for which the team won the Nobel Prize in Chemistry in 2024! Ironwood, the latest TPU that Google unveiled in April this year,

makes it possible to combine 9216 chips into a single pod enabling 42.5 Exaflops of compute, while doubling the PPW of its previous version Trillium.

Amazon's Trainium and Inferentia chips: As the names suggest, Trainium is designed for training AI models, while Inferentia is for deep learning and GenAI inference applications. First- and second-generation Inferentia chips are used on Amazon Elastic Compute Cloud (Amazon EC2), and Amazon Web Services (AWS) claims that they deliver 2x and 4x the throughput of comparable EC2 instances, with lower latency and costs. Case studies indicate that customers such as Finch AI, Sprinklr, Money Forward, Amazon Alexa, Leonardo.ai, Deutsche Telekom, and Qualtrics have realised 70-90% cost savings with Inferentia. The AWS Neuron SDK helps developers deploy models on the AWS Inferentia chips and train them on AWS Trainium chips. It integrates natively with popular frameworks, so developers can run their existing code and workflows on Inferentia chips.

Meta's MTIA: The second generation of the Meta Training and Inference Accelerator (MTIA 2i) is now deployed at scale, and the company claims in a whitepaper that it reduces their total cost of ownership (TCO) by